

Topic 04: Separable Equations & Initial Value Problems (The Clean Split)

Professor Leonard Master Series · Lessons 12 to 14

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: The Great Divorce

Separable differential equations are the bread and butter of calculus. Professor Leonard calls this "**The Great Divorce**":

"You have a crowded room where x 's and y 's are mingled together. Your job is to put all the y 's on the left side of the room with dy , and all the x 's on the right side with dx . Once they are completely separated, both sides can be integrated independently!"

$$\frac{dy}{dx} = g(x)h(y) \implies \frac{1}{h(y)} dy = g(x)dx$$

2. Step-by-Step Whiteboard Walkthrough

Problem: Solve the Initial-Value Problem:

$$\frac{dy}{dx} = \frac{2x(y^2 + 1)}{y}, \quad y(0) = 1$$

Step 1: Separate the variables

Move y and $(y^2 + 1)$ to the left, dx and x to the right:

$$\frac{y}{y^2 + 1} dy = 2x dx$$

Step 2: Integrate both sides

$$\int \frac{y}{y^2 + 1} dy = \int 2x dx$$

- **Left Side:** Use u -substitution ($u = y^2 + 1, du = 2y dy \implies y dy = \frac{1}{2} du$):

$$\frac{1}{2} \ln |y^2 + 1| = \frac{1}{2} \ln(y^2 + 1) \quad (\text{since } y^2 + 1 > 0)$$

- **Right Side:**

$$\int 2x dx = x^2 + C$$

Step 3: Combine and isolate y

$$\frac{1}{2} \ln(y^2 + 1) = x^2 + C$$

Multiply by 2:

$$\ln(y^2 + 1) = 2x^2 + 2C$$

Exponentiate both sides (Leonard's Rule: $e^{2x^2+2C} = e^{2C} e^{2x^2} = Ae^{2x^2}$):

$$y^2 + 1 = Ae^{2x^2} \implies y^2 = Ae^{2x^2} - 1$$

Step 4: Apply the initial condition $y(0) = 1$

$$1^2 = Ae^0 - 1 \implies 1 = A - 1 \implies A = 2$$

$$y^2 = 2e^{2x^2} - 1$$

Since $y(0) = 1 > 0$, choose the positive branch:

$$y(x) = \sqrt{2e^{2x^2} - 1}$$

3. Leonard's Red Flag: The "Lost" Equilibrium Solutions

Whenever you divide by a function of y (say, dividing by $h(y)$), you are assuming $h(y) \neq 0$. If there is a constant $y = k$ such that $h(k) = 0$, that constant function is an **Equilibrium Solution**!

- In $\frac{dy}{dx} = y(y - 3)$, dividing by $y(y - 3)$ risks losing $y = 0$ and $y = 3$.
- Always write:

"Check singular solutions: $y = 0$ and $y = 3$."

on the side of your board!